White Team (8:00PM) Intrusion Report
Team 10
StAABility

Table of Contents

IDS Activity Overview 3
FTP 7
Remote Operations Server 8
Domain Controller 9
IDS 11
HMI 12

IDS Activity Overview

Nothing has been found

Incident1 – Forged Ticket Flag Exploit

System: Tickets Server (tickets.team10.isucdc.com — 6.87.159.20) **Service:** Ticketing API (HTTP, Port 5000) **Severity:** High

What Red Team Exploited

The ticket validation logic contained a hardcoded backdoor condition inside the application source code. Specifically, the validation function checked whether a ticket's expiration timestamp (exp) was greater than or equal to 2000000000 — a Unix epoch value corresponding to May 18, 2033. If this condition evaluated to true, the application immediately returned a hardcoded flag string embedded directly in the code, completely bypassing all other validation checks. No authentication, no database query, and no elevated privilege was required to reach this branch — it was reachable by any client that could submit a correctly structured request to the API on port 5000.

The vulnerability had three underlying root causes: a hardcoded conditional branch in production validation code that should not exist; the complete absence of any upper bound check on the exp field, allowing arbitrarily large timestamp values; and an insufficiently restricted certificate trust model that allowed externally crafted tickets to be accepted as valid by the server.

How Red Team Did It

Red Team's approach was straightforward and required no network intrusion or privilege escalation. They crafted a ticket with an expiration value of exp = 2000000001, placing it just above the trigger threshold. They then signed the ticket using a certificate that fell within the machine's trusted certificate store. Once signed, they submitted the ticket to the validation endpoint on port 5000.

The server accepted the signature as valid, evaluated the expiration condition, found it to be true, and returned the embedded forged flag directly in the API response. The entire exploit operated purely at the application layer — a business logic attack that took advantage of a single flawed conditional, requiring nothing more than a crafted ticket and a trusted certificate to execute successfully.

Incident 2
On February 21, 2026, at approximately 05:56 AM (server time), user melissa.robles successfully authenticated to the server mgmt.team10.isucdc.com. . Following authentication, the user executed the command /bin/su and successfully escalated privileges to the root account. This privilege escalation appears to have been possible due to a system misconfiguration allowing unauthorized or unintended access to root. Once root access was obtained, the user had full administrative control over the system. At approximately 05:57 AM, a password change was recorded for the account cdc. This activity should be reviewed to determine whether it was authorized administrative maintenance or the result of a security misconfiguration leading to unintended privilege escalation

the logs: https://imgur.com/a/vJy4XUo

Incident 3

These flags has been found at the following location: G:\AD C:\Windows\System32. The search has been completed successfully and the flag has been identified by Red. Please proceed with the next steps accordingly.

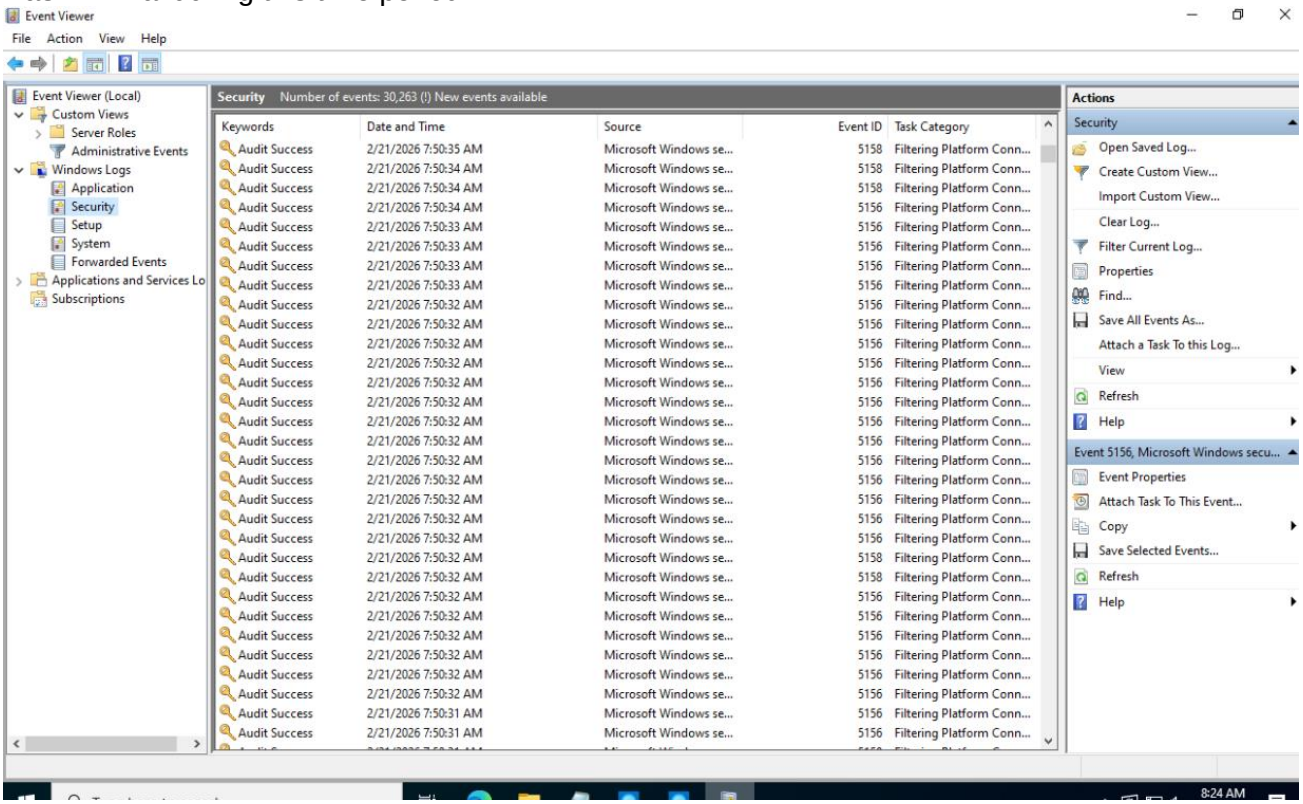
The logs: https://imgur.com/a/AANyEb7

FTP

There were not any malicious login attempts for FTP. The following auth log failures resulted from admins typing their password incorrectly.

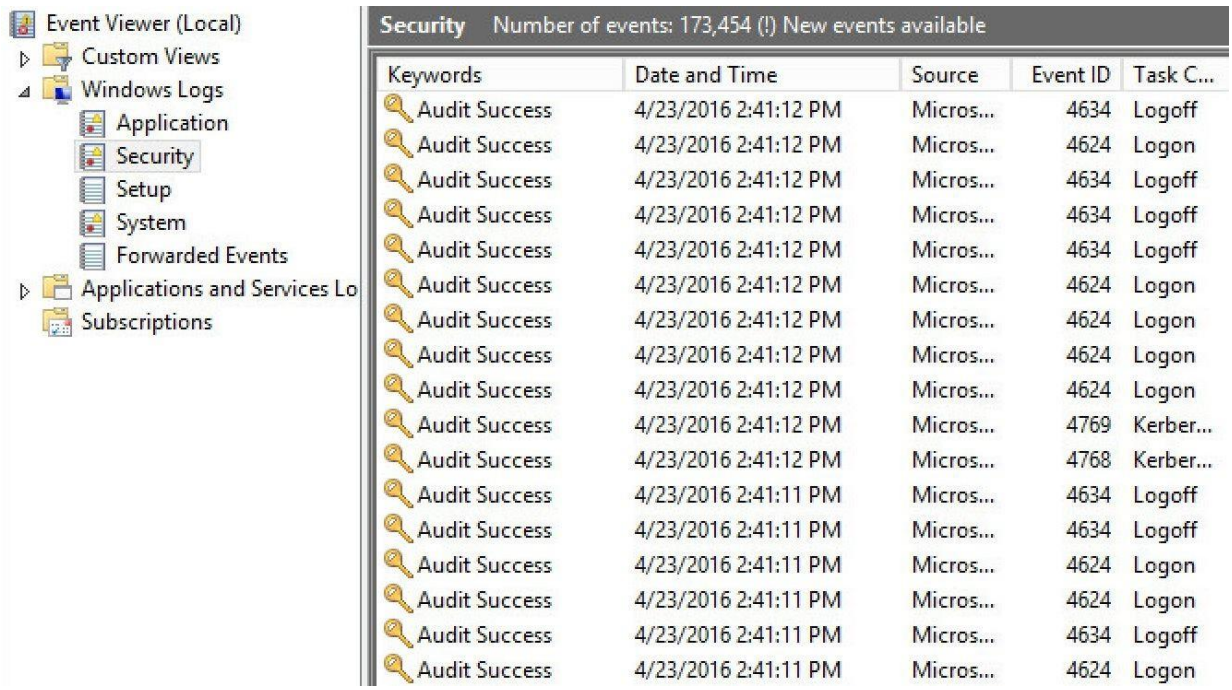
Remote Operations Server

Same as before, we haven't noticed any strange behavior according to the logs below. Login activity was minimal during this time period.



Domain Controller

Same as before, we haven't noticed any strange behavior according to the logs below. Login activity



Keywords	Date and Time	Source	Event ID	Task C...
Audit Success	4/23/2016 2:41:12 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:12 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:12 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4769	Kerber...
Audit Success	4/23/2016 2:41:12 PM	Micros...	4768	Kerber...
Audit Success	4/23/2016 2:41:11 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:11 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:11 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:11 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:11 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:11 PM	Micros...	4624	Logon

was minimal during this time period

IDS

Our IDS does not appear to be compromised. As before, its location in our network makes it difficult for intruders to access it

HMI

Our HMI does not appear to be compromised.